

OUTPUT 1: Intelligent Email and Password Validator using Regular Expressions

```
● PS C:\COURSE\DSA0302\CO1_AT3> & "C:\Users\Shivaji V\AppData\Local\Microsoft\Windows\Apps\PlatformToolbox\0302/CO1_AT3/NLP 1.py"
Enter Email: shivaji.15@gmail.com
Enter Password: ShivaJI.VasanTH@192424370
Enter Mobile Number: 9876543210
Valid Email
Strong Password
Valid Mobile Number
```

OUTPUT 2: Finite State Automata (FSA) Simulator

```
● PS C:\COURSE\DSA0302\CO1_AT3> & "C:\Users\Shivaji V\AppData\Local\Microsoft\Windows\Apps\PlatformToolbox\0302/CO1_AT3/NLP 2.py"
Enter number of strings: 3
Enter string: abaab

Input : abaab
Transition Path : q0 → q1 → q2 → q1 → q1 → q2
Accepted
Enter string: abab

Input : abab
Transition Path : q0 → q1 → q2 → q1 → q2
Accepted
Enter string: aaaa

Input : aaaa
Transition Path : q0 → q1 → q1 → q1 → q1
Rejected
```

OUTPUT 3: Pattern Matching Engine using Regular Expressions

```
= RESTART: C:\COURSE\DSA0302\CO1_AT3\
Text:
Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 1
Dates: ['12/09/2026']

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 2
Phone Numbers: ['9876543210']

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 3
Hashtags: ['#NLP']

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 4
Mentions: ['@OpenAI']

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 5
Enter prefix: process
Matching Words: ['processing']

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 6
Enter suffix: es
Matching Words: []

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 7
Enter word: language
Word Found

Menu
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Quit

Enter your choice: 8
Program Ended.
```